ENGINEERING MONOGRAPH

TURBINES AND COMPRESSORS

Fundamentals, Thermodynamics, Types, Applications and Operation

Energy Conversion Machines for Power Generation and Process Industries

An introductory engineering monograph

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1. Introduction

Turbines and compressors are the two most important classes of turbomachines — devices that transfer energy between a rotating shaft and a fluid. Although they perform opposite functions, they share the same underlying physics: both impart or extract energy from a fluid through the dynamic action of blades on a rotating rotor.

A **turbine** converts the energy stored in a fluid (pressure, kinetic, or potential energy) into useful mechanical work that drives a generator, pump, or propeller. A **compressor** does the reverse: it consumes mechanical work to raise the pressure of a gas. Together they form the backbone of modern power generation, aviation, oil and gas processing, refrigeration, and chemical manufacturing.

This monograph presents the operating principles, thermodynamic foundations, principal types, and practical considerations for both machine families, and concludes with a comparison of their characteristics.

1.1 Why these machines matter

- Roughly 80% of the world's electricity is generated by turbines of one form or another.
- Compressors consume about 10% of all industrial electricity, making them a major target for efficiency improvement.
- Gas turbines and compressors are essential to jet propulsion, pipeline transport, and LNG production.

2. Fundamentals of Turbomachinery

2.1 Classification

Turbomachines are classified by the direction of energy transfer and by the path of the fluid:

- **Turbines** — fluid energy → shaft work (steam, gas, water, wind turbines).
- **Compressors / pumps** — shaft work → fluid energy (centrifugal, axial, reciprocating, screw machines).
- **Impulse machines** — energy transfer by change in momentum of the fluid (e.g., Pelton wheel, pressure-compounded steam turbines).
- **Reaction machines** — energy transfer by pressure change across the moving blades (e.g., most gas turbines, Francis hydraulic turbines).

2.2 The Euler turbomachine equation

The theoretical specific work exchanged between rotor and fluid is given by the Euler equation:

$$W = U_2 C_{\theta 2} - U_1 C_{\theta 1}$$

where U is the blade peripheral speed and C_θ is the tangential (whirl) component of absolute fluid velocity at rotor inlet (1) and outlet (2). This single relation explains most of the design choices in turbine and compressor engineering: tip speed limits head capability, and blade angles determine the energy exchange per stage.

2.3 Degree of reaction

The degree of reaction R measures how much of the total enthalpy drop (or rise) occurs in the rotor rather than in the stator. A 50% reaction design splits the pressure change evenly between moving and fixed blades and is the most common compromise for axial-flow machines, balancing efficiency and mechanical loading.

3. Turbines

3.1 Steam turbines

Steam turbines expand high-pressure, high-temperature steam through successive rows of stationary nozzles and moving blades. They dominate thermal power plants (coal, nuclear, concentrated solar) and drive large mechanical loads in refineries and petrochemical plants.

- **Impulse (Curtis / Rateau)** — pressure drop occurs entirely in nozzles; blades receive a momentum impulse. Robust, used for control stages.
- **Reaction (Parsons)** — pressure drops across both fixed and moving blades; higher efficiency, tighter clearances.
- Modern units reach inlet conditions above 25 MPa / 600 °C with condenser pressures near 5 kPa, giving thermal efficiencies up to 48% (ultra-supercritical plants).

3.2 Gas turbines

Gas turbines compress ambient air (pressure ratio 15–40 in aero engines, up to 45 in industrial units), burn fuel in a combustion chamber, and expand the hot gas through a turbine. The turbine produces more power than the compressor consumes, and the surplus drives a generator or a propeller.

- Simple-cycle efficiency: roughly 35–44%.
- Combined-cycle (gas turbine + steam turbine sharing a heat-recovery boiler): efficiencies above 64%, the highest of any thermal plant.
- Turbine inlet temperatures exceed 1600 °C in modern machines, requiring single-crystal superalloy blades with internal cooling channels and thermal-barrier coatings.

3.3 Hydraulic turbines

- **Pelton (impulse)** — for high heads (250–1000+ m); free jets strike double-cup buckets.
- **Francis (mixed flow)** — for medium heads (40–600 m); the most common design worldwide.
- **Kaplan (axial, adjustable blades)** — for low heads (< 40 m) and high flows, e.g., run-of-river stations.

3.4 Wind turbines

Modern horizontal-axis wind turbines convert the kinetic energy of moving air into electricity. Power scales with the cube of wind speed and the square of rotor diameter — hence the growth of machines exceeding 15 MW offshore with rotors over 240 m.

4. Thermodynamic Cycles of Turbines

4.1 Rankine cycle (steam)

The ideal Rankine cycle consists of four processes: isentropic pumping of feedwater, constant-pressure heat addition in a boiler, isentropic expansion in the turbine, and constant-pressure heat rejection in the condenser. The thermal efficiency is:

$$\eta = (W_{turbine} - W_{pump}) / Q_{in} \approx (h_1 - h_2) / (h_1 - h_4)$$

Reheat (expanding in two stages with re-superheating between them) and regeneration (feedwater preheating with extracted steam) materially raise efficiency and reduce turbine exhaust wetness.

4.2 Brayton cycle (gas turbine)

The ideal Brayton cycle compresses air isentropically, adds heat at constant pressure, expands isentropically, and rejects heat at constant pressure. Back-work ratio — the fraction of turbine work consumed by the compressor — is 40–50%, much higher than the 1–2% of a Rankine pump, so compressor efficiency is critical to gas-turbine performance.

Intercooling between compressor stages and reheating between turbine stages improve specific output; the exhaust heat feeds a steam cycle in combined-cycle plants.

4.3 Losses and real-machine efficiency

- Profile loss (boundary layers on blade surfaces), secondary loss (endwall vortices), and tip-leakage loss.
- Real stage isentropic efficiencies: 85–93% for well-designed axial stages.
- Condensing steam turbines are limited by wetness at the last stages; moisture separators and blade heating mitigate erosion.

5. Compressors

5.1 Positive-displacement types

- **Reciprocating (piston)** — gas is trapped in a cylinder and compressed by a piston. Capable of very high pressures (up to 500+ bar); used in gas pipelines, CNG stations, and refrigeration. Low flow, pulsating delivery.
- **Screw (rotary twin-lobe)** — two intermeshing rotors trap and squeeze the gas axially. Oil-injected versions are compact and quiet, dominating industrial air supplies up to ~25 bar.
- **Scroll / vane / roots blowers** — smooth or low-pressure delivery for HVAC, supercharging, and pneumatic conveying.

5.2 Dynamic types

- **Centrifugal** — gas enters the impeller axially, is whirled outward by radial blades, and leaves with high velocity that is converted to pressure in a diffuser/volute. Continuous flow, few moving parts, ideal for large volumes at moderate pressure (3–12 stages reach 100+ bar).
- **Axial** — many rows of aerofoil blades, flow parallel to the shaft. High efficiency (88–92% polytropic), very high flow rates; used in jet engines, large air separation plants, and LNG trains.

5.3 Thermodynamics of compression

Unlike turbines, compression work rises steeply with temperature. Isothermal compression (with perfect intercooling) requires the least work; real polytropic compression lies between isothermal and adiabatic:

$$W_{poly} = n/(n–1) \cdot p_1V_1 \cdot [(p_2/p_1)^{(n–1)/n} – 1]$$

Multi-staging with intercooling moves the process toward the isothermal ideal and is standard practice whenever overall pressure ratios exceed about 4–5 per stage.

6. Turbines versus Compressors — A Comparison

The two machine families are thermodynamic mirror images, and many design lessons are shared. The table below summarises the key contrasts.

Aspect	Turbine	Compressor
Energy flow	Fluid → shaft	Shaft → fluid
Fluid behaviour	Expands, cools, accelerates	Compressed, heats up, decelerates
Blade loading	Favourable pressure gradient; flow stays attached	Adverse pressure gradient; prone to stall and surge
Typical stage count	Few stages, large enthalpy drop each	Many stages, small pressure rise each
Key limit	Blade temperature / creep strength	Surge margin, choke (Mach limits)
Efficiency definition	Isentropic: actual work / ideal work	Isentropic/polytropic: ideal work / actual work
Representative types	Steam, gas, hydraulic, wind	Reciprocating, screw, centrifugal, axial

Table 1 — Key differences between turbine and compressor design.

6.1 Stall, surge, and choke

A compressor stage can stall when the incidence angle on its blades becomes too large, causing boundary-layer separation. System-wide instability — **surge** — is a violent flow reversal that can destroy a machine in seconds and must be prevented with anti-surge recycle valves and fast controls. **Choke** (sonic velocity in the blade passages) sets the maximum flow. Turbines face no surge analogue, which is one reason compressor operation demands more instrumentation and protection logic.

7. Applications

7.1 Power generation

- Coal, nuclear, and geothermal plants — steam turbines.
- Peaking plants and combined-cycle stations — gas turbines plus steam turbines.
- Hydroelectric dams and pumped-storage schemes — Francis, Kaplan, and Pelton units.
- Onshore and offshore wind farms — horizontal-axis wind turbines.

7.2 Oil, gas, and petrochemicals

- Pipeline compressors (centrifugal, driven by gas turbines or motors) move natural gas across continents.
- Refrigeration and process compressors in LNG plants, refineries, and air separation.
- Steam turbines drive large pumps, blowers, and compressors as mechanical drives where electricity is impractical.

7.3 Aviation

The turbofan engine is an extreme integration of both machine types: a multistage axial compressor (up to 40:1 pressure ratio) feeding an annular combustor, with the turbine extracting just enough work to drive the fan and compressor while the remainder leaves as jet thrust.

7.4 Everyday life

- Refrigerators, air conditioners, and heat pumps — hermetic reciprocating, scroll, and rotary compressors.
- Turbochargers on diesel and petrol engines — exhaust-gas turbine driving a centrifugal compressor.
- Dental drills, paint sprayers, and workshop tools — small screw and vane compressors.

8. Operation, Maintenance, and Trends

8.1 Monitoring and condition assessment

- Vibration analysis (unbalance, misalignment, bearing wear) and rotor orbit tracking.
- Thermal-performance monitoring: isentropic efficiency degradation signals fouling or leakage.
- Oil analysis for bearings and gearboxes; blade-tip clearance and thermographic inspections.

8.2 Common failure modes

- Turbine blades — creep, fatigue, erosion/corrosion, foreign-object damage.
- Compressor blades — fouling (efficiency loss of 2–5% per year without washing), surge damage, rubs.
- Seals and bearings — a large share of unplanned outages in both families.

8.3 Future directions

- Hydrogen-ready gas turbines and supercritical-CO_2 power cycles.
- Larger offshore wind turbines, floating platforms, and airborne turbines.
- Additive-manufactured blades with internal cooling lattices; magnetic bearings and dry gas seals.
- Digital twins and AI-driven predictive maintenance for compressor fleets and turbine plants.

9. Conclusion

Turbines and compressors embody the same turbomachinery physics applied in opposite directions. Mastery of the Euler equation, reaction, and the thermodynamics of expansion and compression allows engineers to build machines that generate most of the world's electricity, move its energy resources, and drive countless industrial processes. As the energy system decarbonises, these machines — burning hydrogen, capturing CO_2, or harvesting wind — will remain central to how civilisation converts and transports energy.